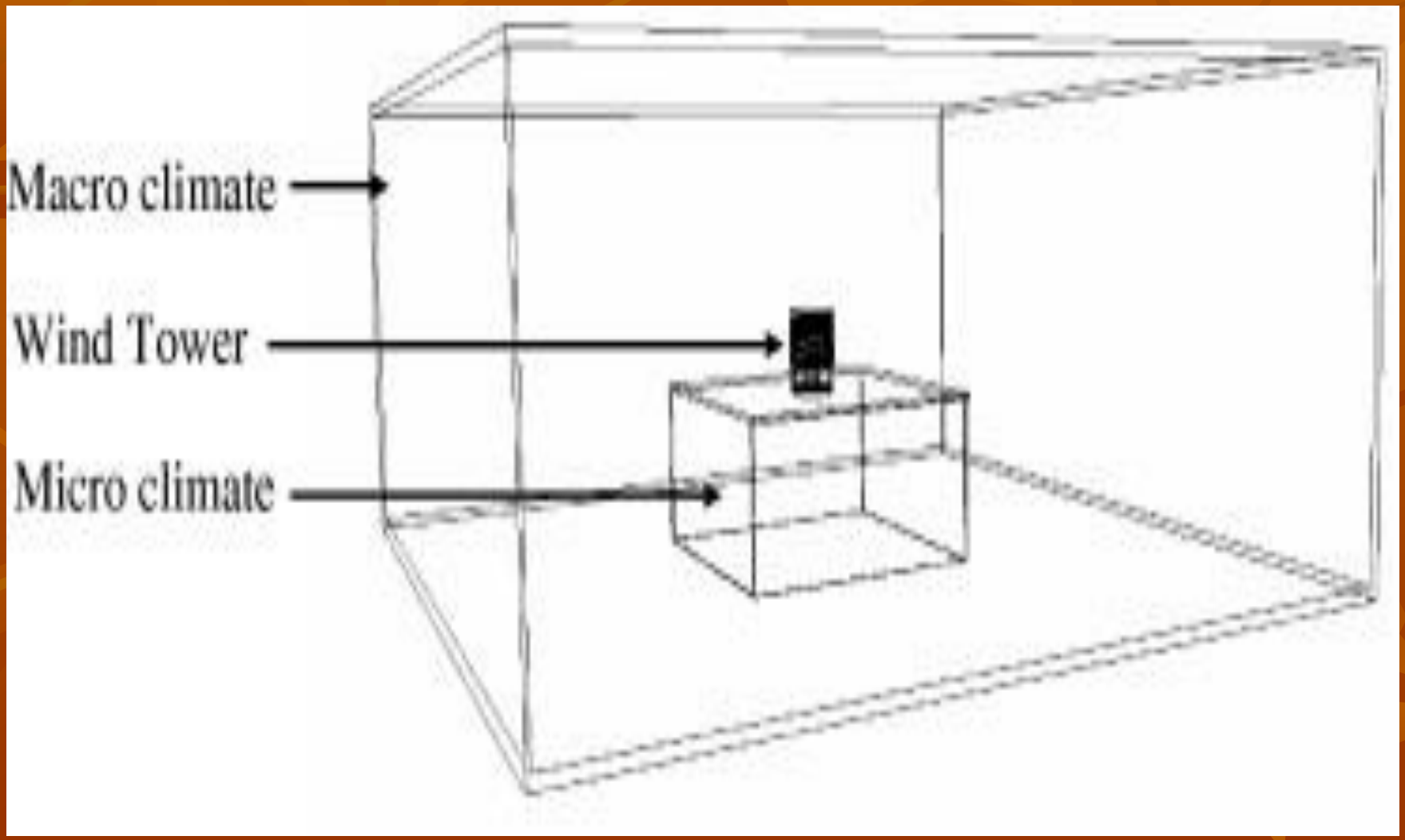
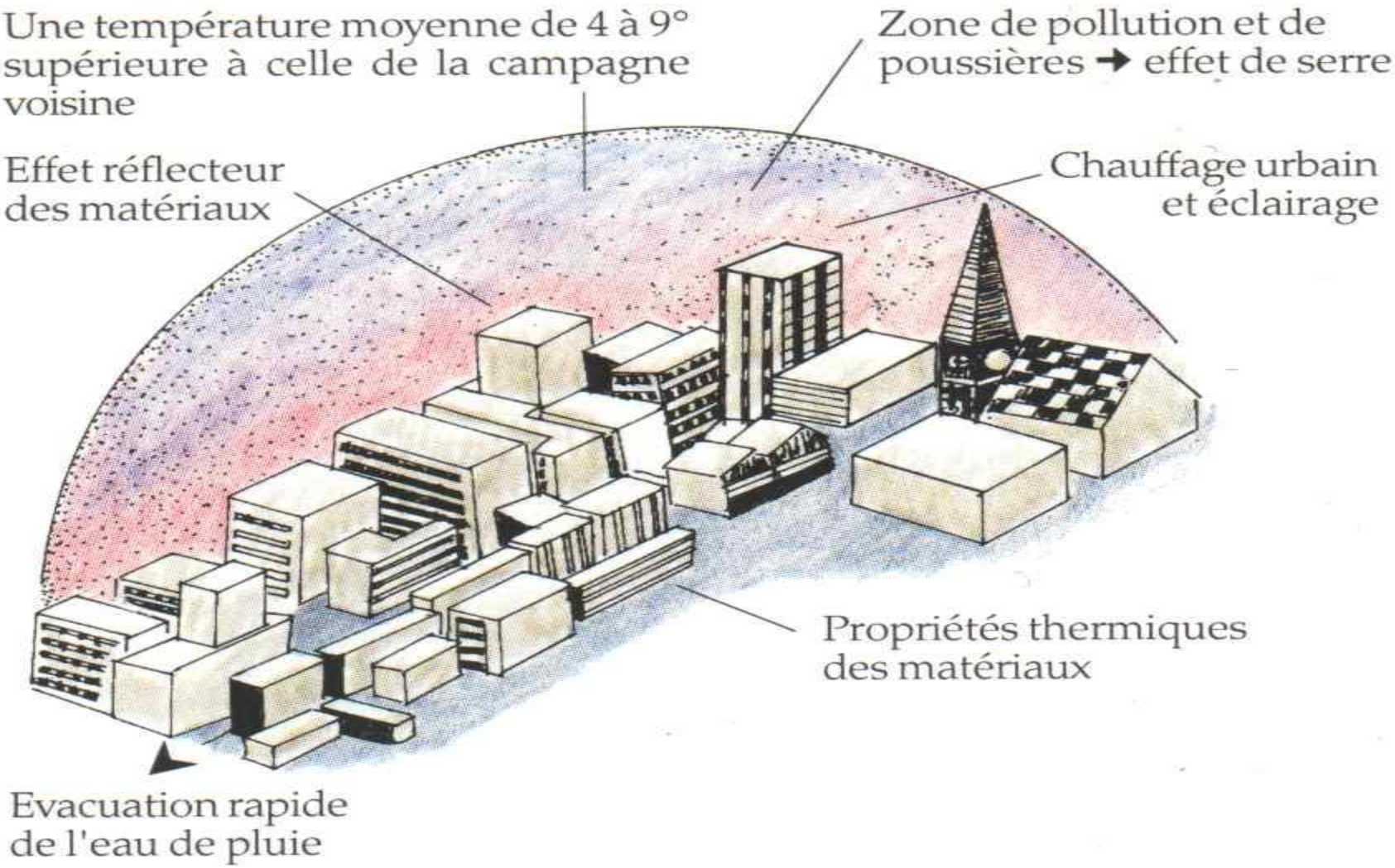
Vi khí hậu

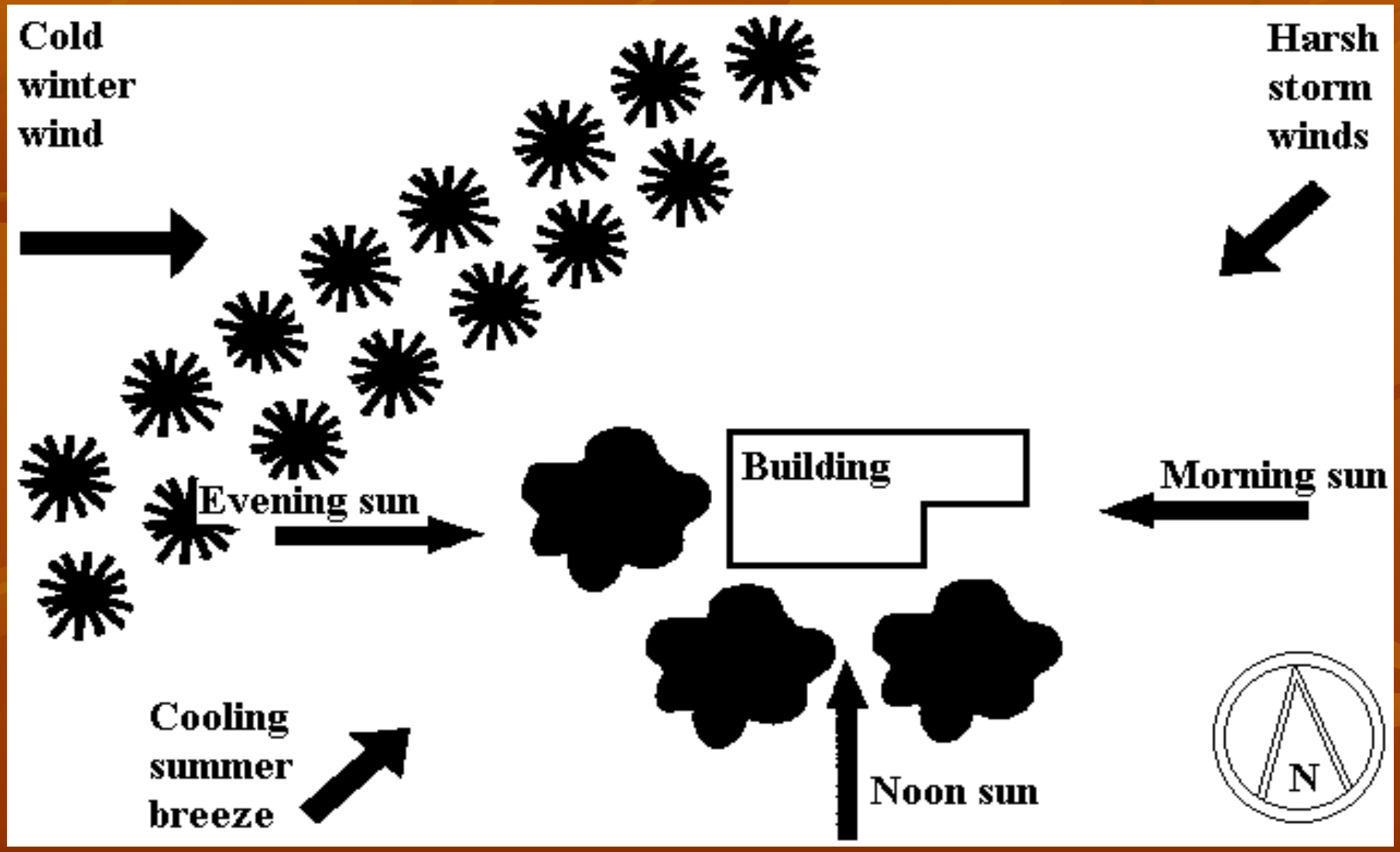
MACRO AND MICRO CLIMATE



MACRO AND MICRO CLIMATE

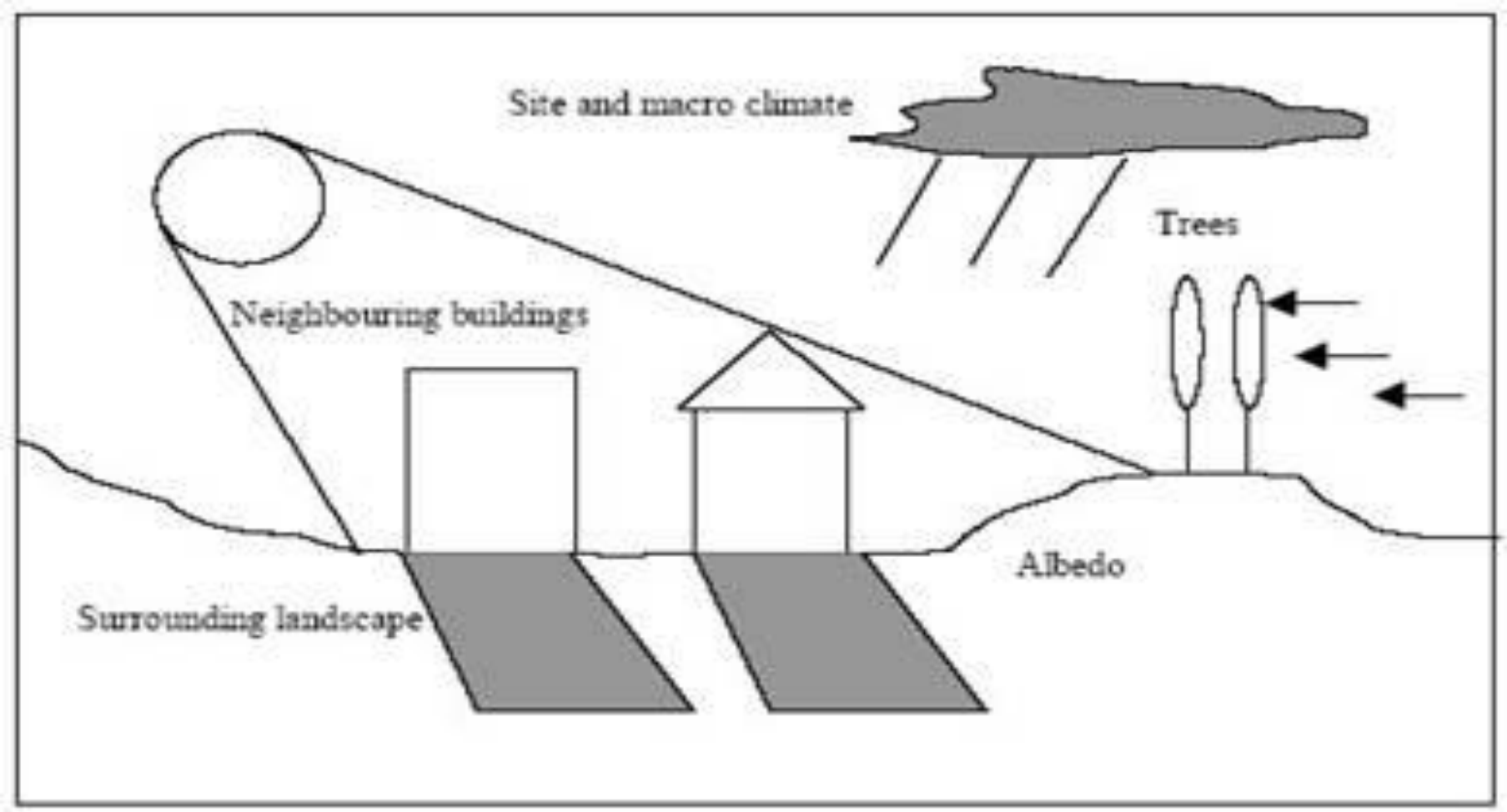


MACRO AND MICRO CLIMATE

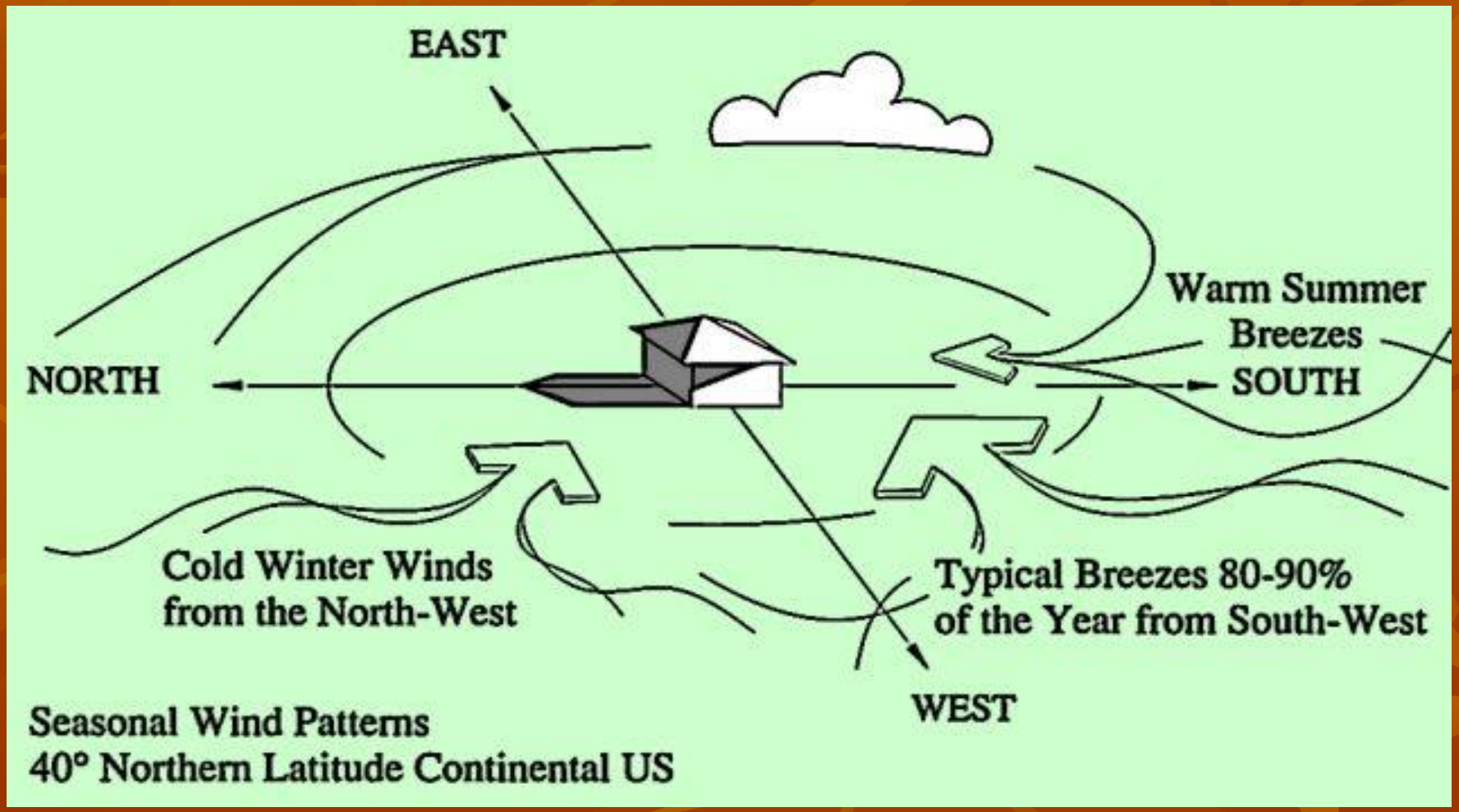


MACRO AND MICRO CLIMATE

MACRO CLIMATE



MACRO CLIMATE



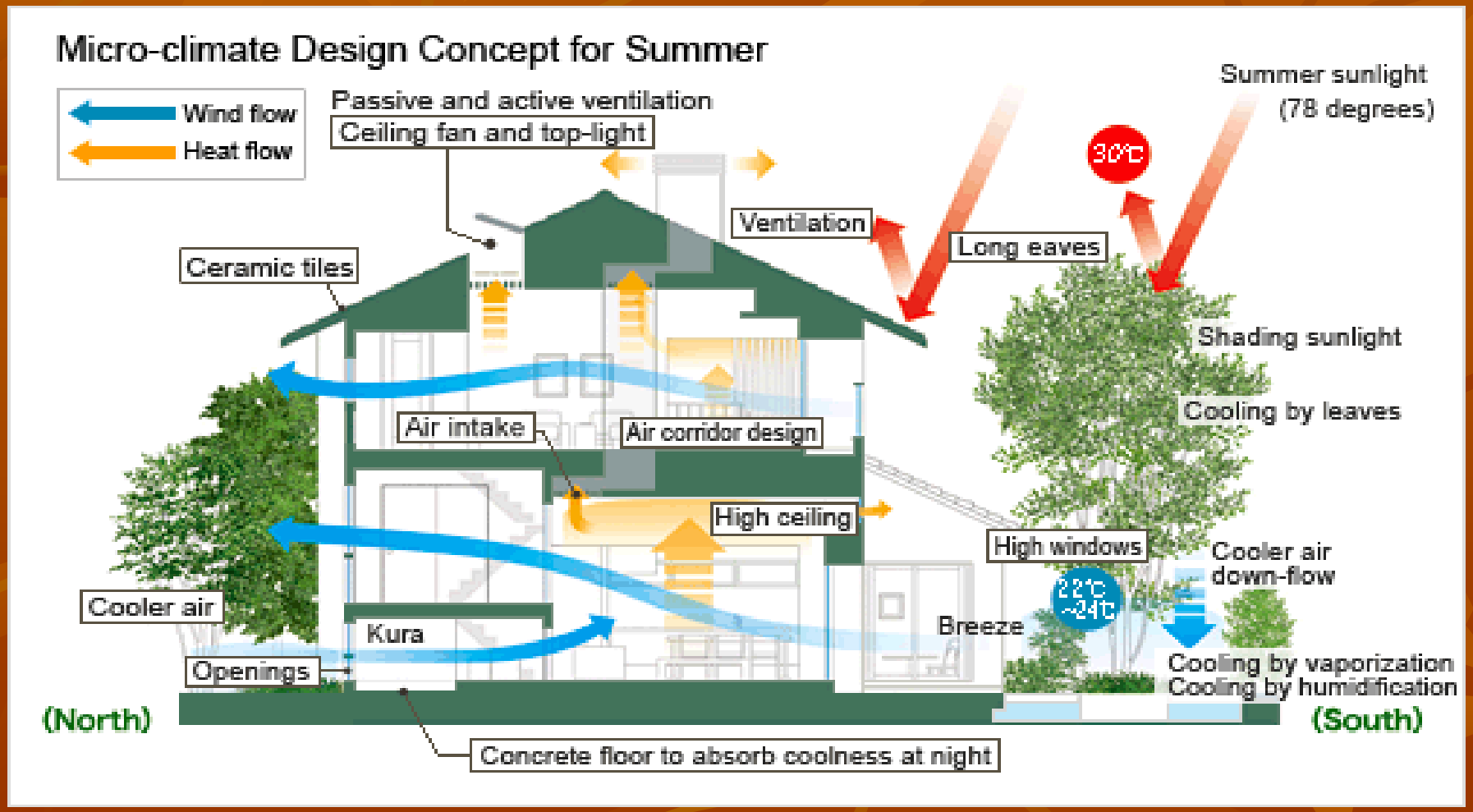
Macroclimate

The name given to describe the prevailing climate of a particular region.

WWW.WINEOPOLIS.com

MICRO CLIMATE

MICRO CLIMATE



Microclimate

- A microclimate is the weather that is found in a relatively small, very local area (e.g. woodland, town on mountain valley)
- What conditions affect weather conditions locally?
- How would the following affect microclimates?
  - Altitude
  - Exposure and shelter
  - Water
  - Buildings
  - Concrete

-Aspect

-Coasts

-Vegetation

What is a microclimate?

- A microclimate is when the climate in a small area is different to the general area around it
- Many different things can cause this
- (Climate is the average weather conditions of a place over a long period of time – remember??!!)

For more detailed evaluation the climate could be divided into:

- Macro-climate – in given geographic zone
  - "Middle-climate" – in a given place
  - "Micro-climate" – in closed rooms

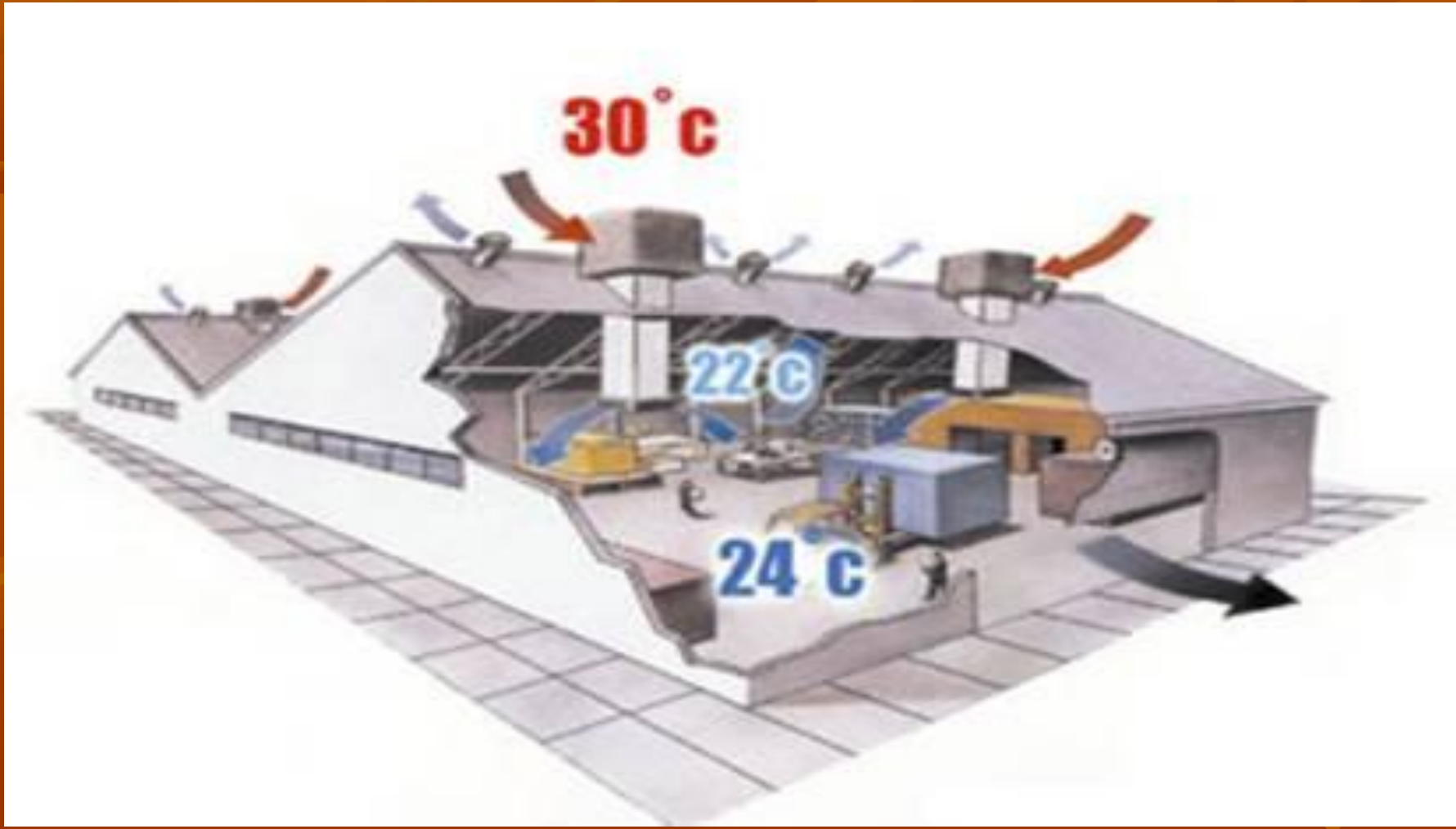
MICRO CLIMATISEUR

MICRO CLIMATISEUR

MICRO CLIMATISEUR

MICRO CLIMATISEUR

MICRO CLIMATISEUR



NHIỆT KẾ

To convert	Use this equation:	Example
Celsius to Fahrenheit °C ⟶ °F	$°F = \left(\frac{9}{5} \times °C\right) + 32$	Convert 45°C to °F. $°F = \left(\frac{9}{5} \times 45°C\right) + 32 = 113°F$
Fahrenheit to Celsius °F ⟶ °C	$°C = \frac{5}{9} \times (°F - 32)$	Convert 68°F to °C. $°C = \frac{5}{9} \times (68°F - 32) = 20°C$
Celsius to Kelvin °C ⟶ K	$K = °C + 273$	Convert 45°C to K. $K = 45°C + 273 = 318\ K$
Kelvin to Celsius K ⟶ °C	$°C = K - 273$	Convert 32 K to °C. $°C = 32\ K - 273 = -241°C$

CONVERTING BETWEEN FAHRENHEIT AND CELSIUS

$$T_{Fahrenheit} = \frac{9}{5} T_{Celsius} + 32$$

$$T_{Celsius} = \frac{5}{9} \left(T_{Fahrenheit} - 32 \right)$$

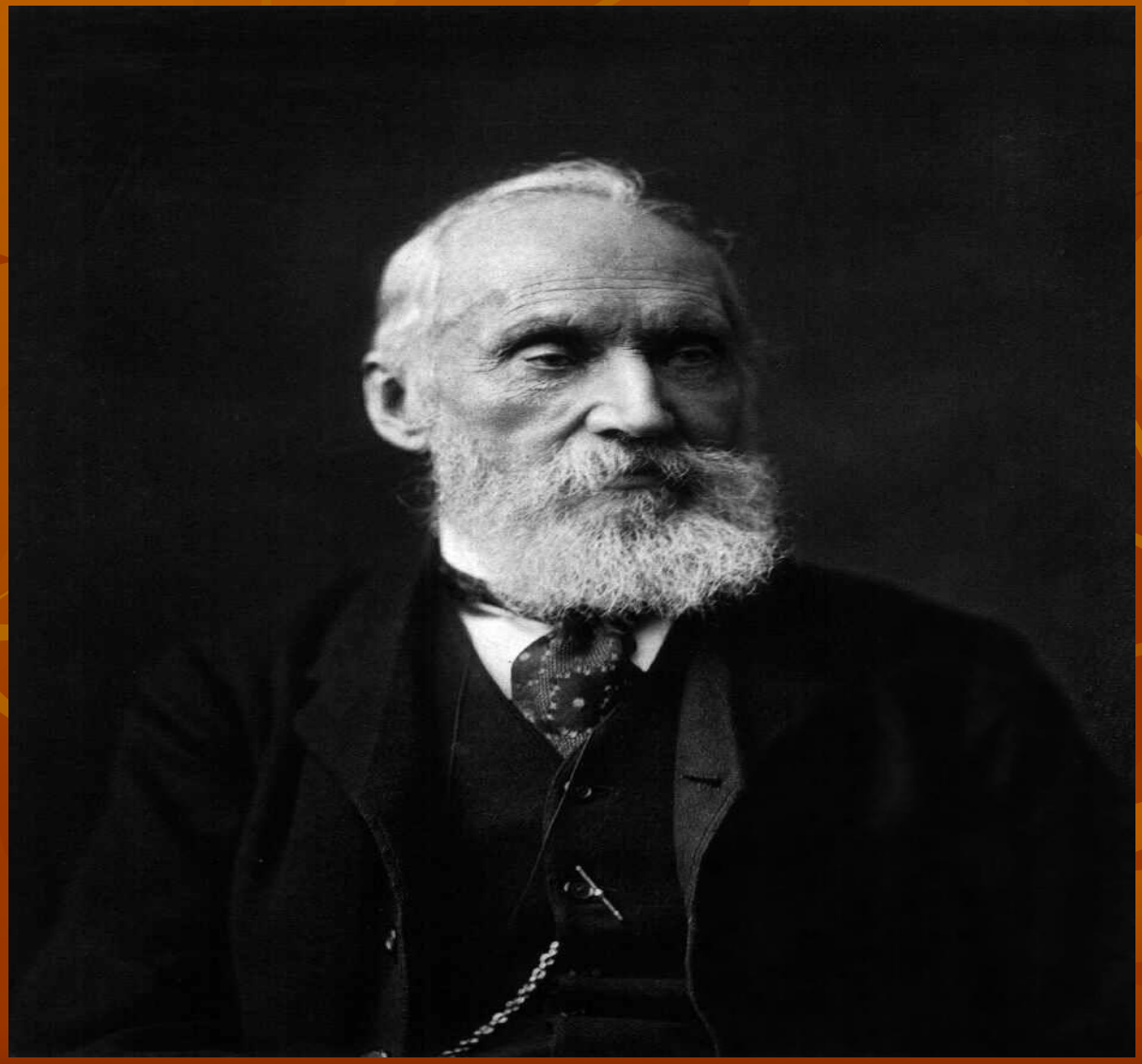
Fahrenheit-Celsius conversion table

°F	°C
105	40.6
104	40.0
103	39.4
102	38.9
101	38.3
100	37.8
99	37.2
98	36.7
97	36.1
96	35.6

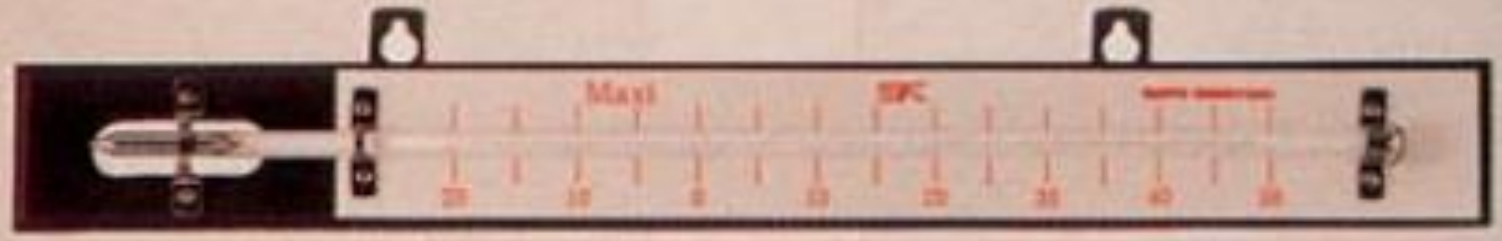
	°F	°C
BOILING POINT*	212°	100°
NORMAL BODY TEMPERATURE	98.6°	37°
ROOM TEMPERATURE	68°	20°
MELTING POINT*	32°	0°

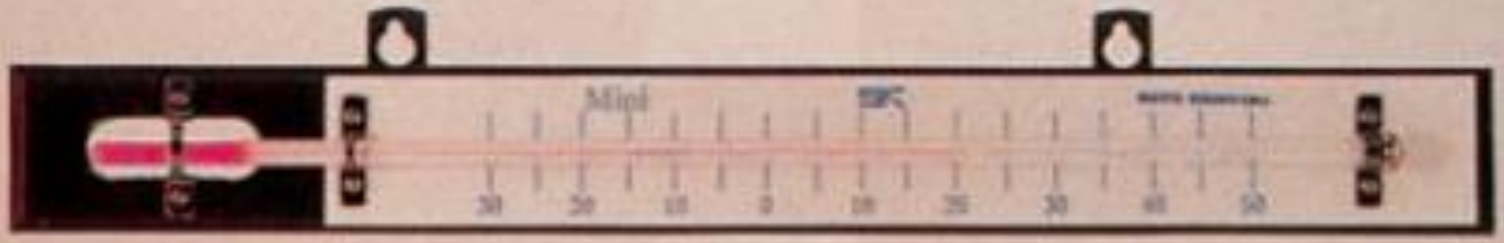
*OF WATER

Kenvin

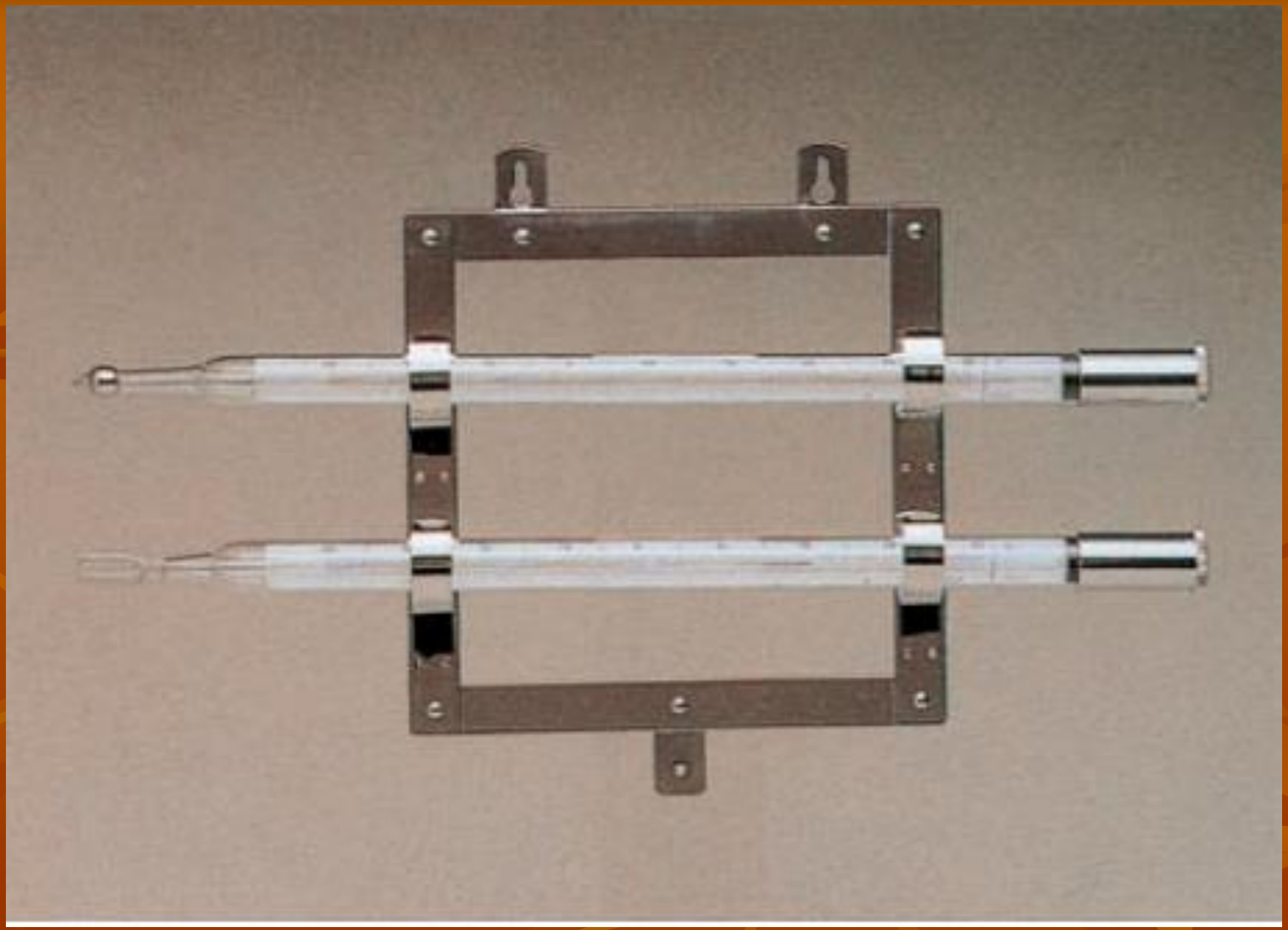


NHIỆT KẾ KIM LOẠI
NHIỆT KẾ ĐIỆN TỬ
36.5
NHIỆT KẾ THỦY NGÂN
NHIỆT KẾ RƯỢU
NHIỆT KẾ


Max


Min

C
F
50
40
30
20
10
0
10
20
30
120
100
80
60
40
20
0
20
superstock.com



MIN
5.4°C
MAX
18.8°C
SATO
RESET
MIN-TEMP-MAX

22.9

TEMPSENS
INSTRUMENTS

23.8

20.5
40.6

TEMPpoint

Max
Min
Set
Esc
Alarm
24.5
54.6
11:54:45
24.12.2009
30
20
5

ẨM KẾ

Lượng hơi nước tối đa trong không khí

Nhiệt độ (0C)	Lượng hơi nước (g/m3)
0	2
10	5
20	17
30	30

No. 1519-00
PSYCHROMETER SK-1

SPECIFICATIONS

Measuring range:	–10 to 50℃
Min. graduation:	1℃
Accuracy:	±1℃ (at 0 to 30℃)
Dimensions:	28 x 260 x 104 mm
Weight :	0.12 kg

No. 7360-00 FOOSE PSYCHROMETER

SPECIFICATIONS

Measuring range:	–30 to +50°C
Min. graduation:	0.2°C
Accuracy:	±0.2°C
Accessories:	Metal frame, conversion table
Dimensions:	440 x 117 mm (including metal frame)
Weight:	0.3 kg (inclusive of metal frame)

No. 7320-00 AUGUST PSYCHROMETER

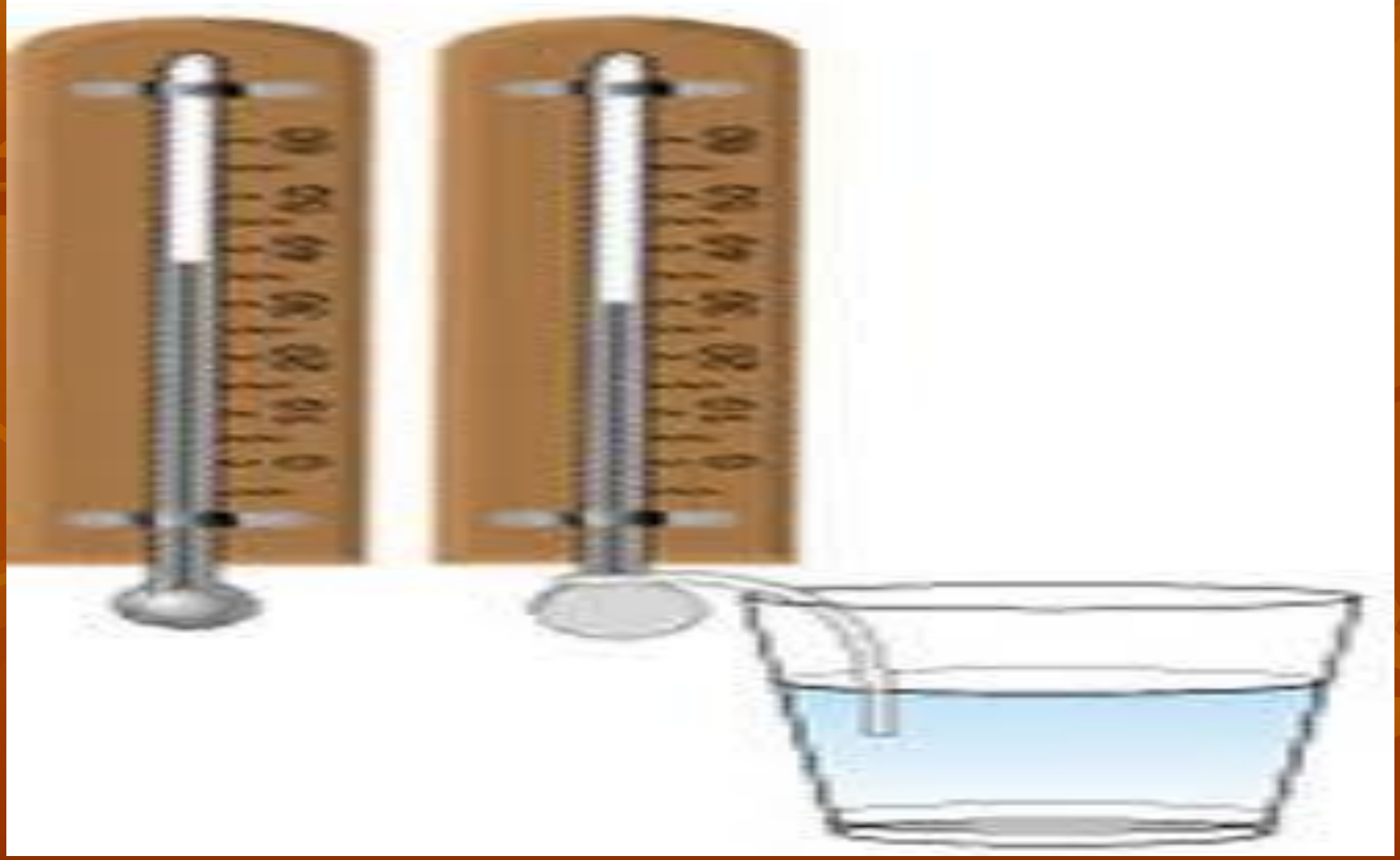
SPECIFICATIONS

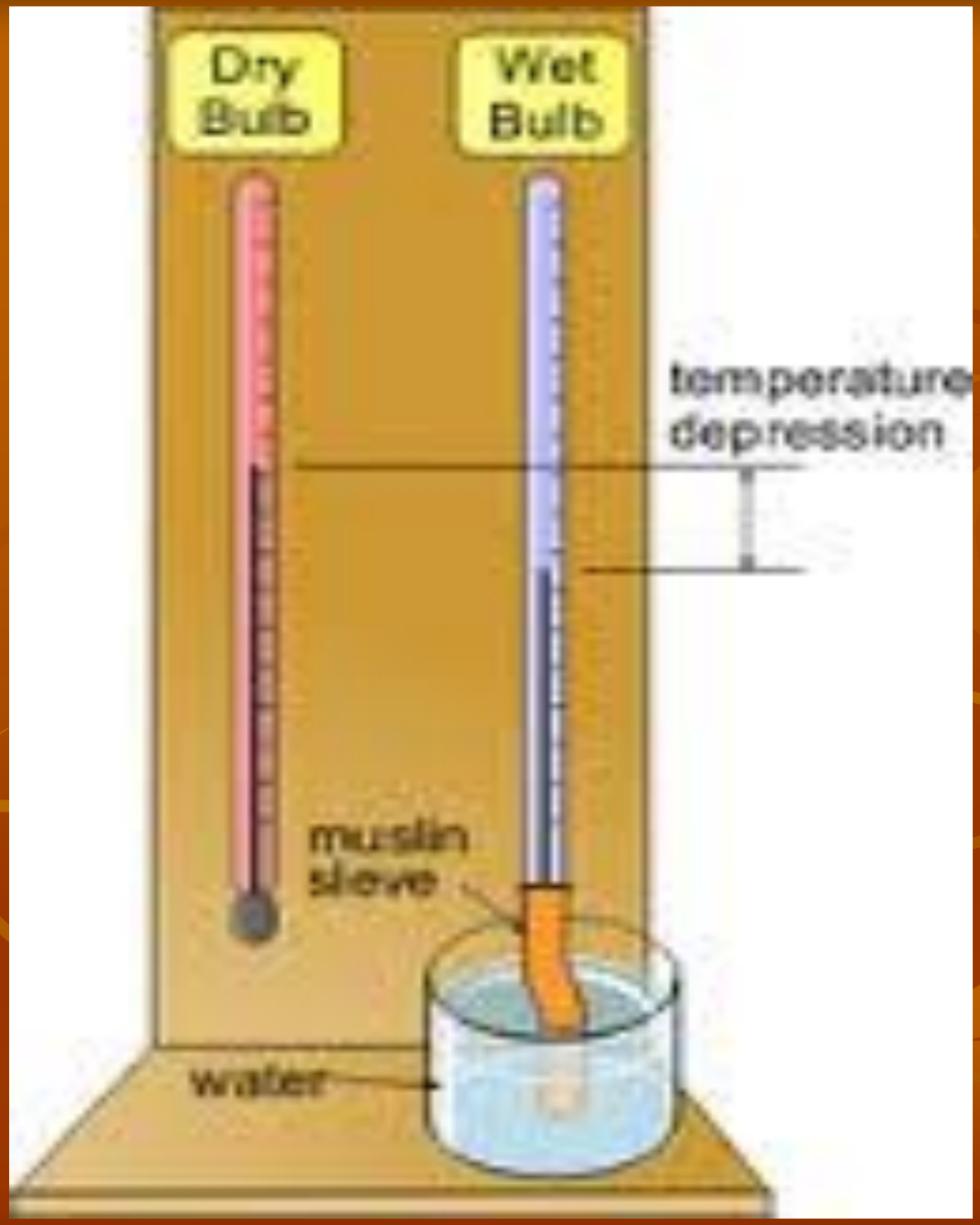
Measuring range:	–20 to +50°C
Min. graduation:	0.5°C
Accuracy:	±0.5°C
Accessories:	Metal frame, conversion table
Dimensions:	420 x 110 mm
Weight:	0.14 kg

AUGUST PSYCHROMETER

AUGUST PSYCHROMETER

AUGUST PSYCHROMETER




Dry Bulb
Wet Bulb
temperature depression
muslin sleeve
water

dreamstime.com

ASSMANN PSYCHROMETER

ASSMANN PSYCHROMETER

ASSMANN PSYCHROMETER

ASSMANN PSYCHROMETER

ASSMANN PSYCHROMETER

ASSMANN PSYCHROMETER

ASSMANN PSYCHROMETER

ASSMANN PSYCHROMETER

No. 7540-00 PRECISION HAIR HYGROTHERMOMETER (HIGHEST I)

SPECIFICATIONS

Sensors:	Human hair for humidity
	Glass thermometer for temperature
Measuring range:	20 to 100% for relative humidity
	−10 to +50°C for temperature
Min. graduation:	1% for relative humidity, 2°C for temperature
Accuracy:	±3% for R.H. at 40 to 80% and 15 to 25°C
	±5% at other than above
	±2°C for temperature
Dimensions:	150 mm dia. x 40 mm
Weight:	Approx. 0.28 kg

No. 7542-00 HAIR HYGROMETER WITH DIGITAL THERMOMETER (HIGHEST II)

SPECIFICATIONS

R. Humidity:	Measuring range:	20 to 100%
	Min. graduation:	2%
	Accuracy:	±4% for R.H. at 40 to 80% and 15 to 25°C
		±7% in other than above
Temperature:	Measuring range:	-20 to +60°C
	Resolution:	digital display every 0.1°C
	Accuracy:	±1°C for temperature at 0 to 40°C
		±2°C for other than above
Dimensions:	135 (D) x 33 (T) mm	
Weight:	0.35 kg	
Accessories:	Bracket: 160 (D) x 4 (T) mm, "AA" size battery x 1	

No. 7562-00 HYGROTHERMOMETER (PALMA II)

SPECIFICATIONS

Sensors:	Synthetic hair for humidity Glass thermometer for temperature
Measuring range:	20 to 99% for relative humidity –10 to +50℃ for temperature
Min. graduation:	1% for relative humidity, 5℃ for tempoerature
Accuracy:	±4% for R.H. at 30 to 80% and 15 to 25℃ ±8% at other than above
Dimensions:	105 mm dia. x 18 mm (D)
Weight:	Approx. 0.1 kg

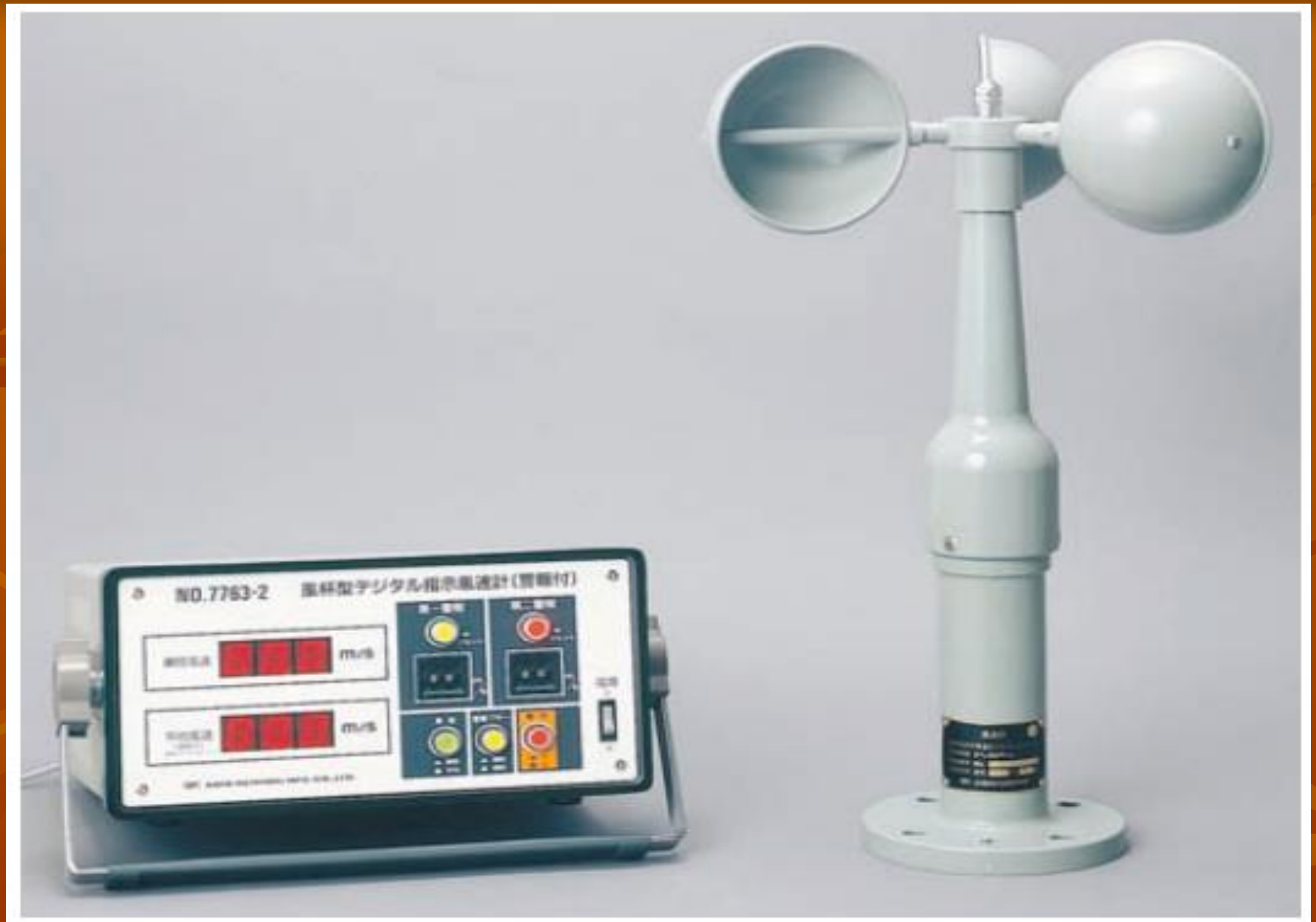
10
20
30
40
50
60
70
80
90
100
%

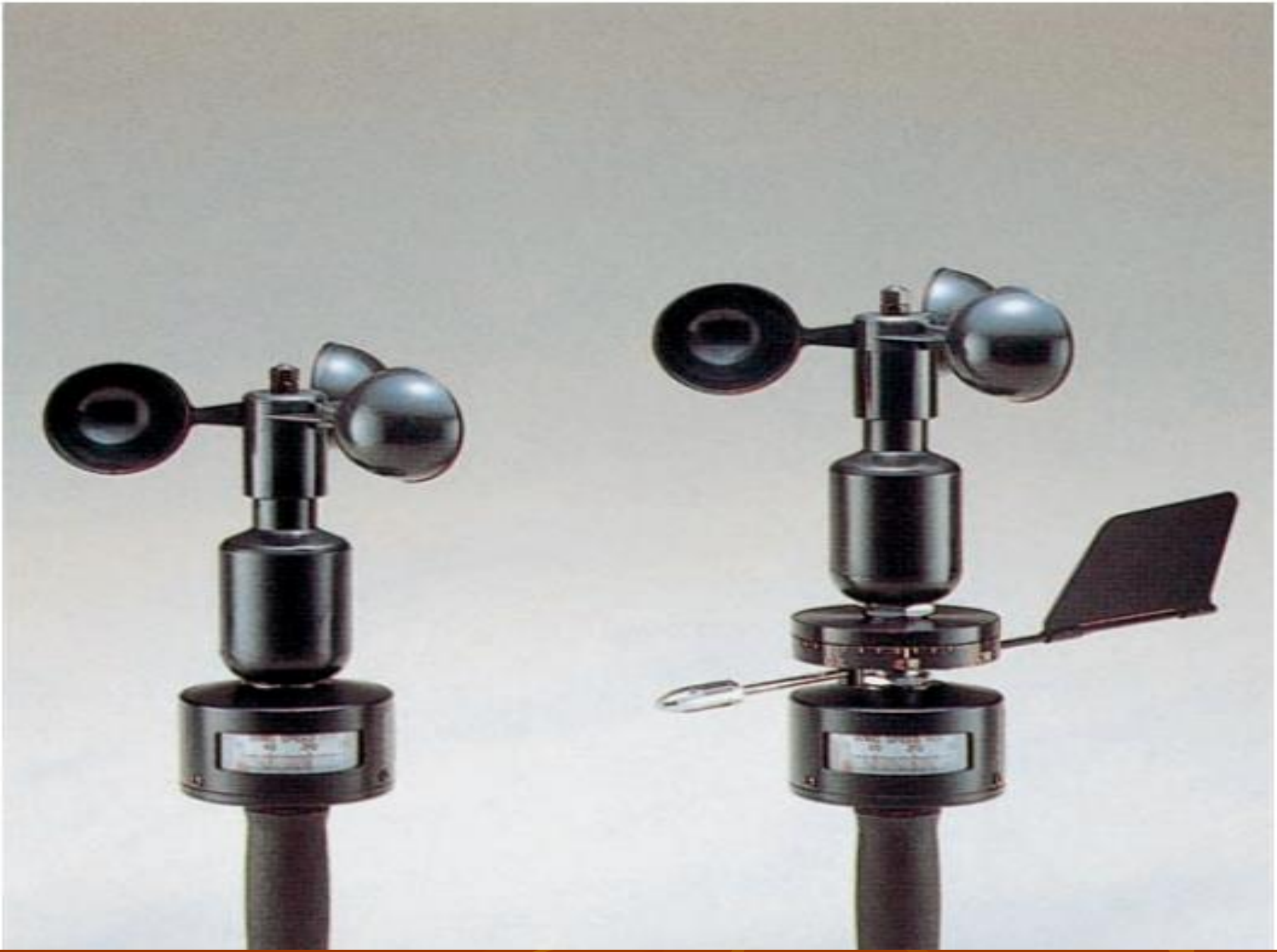
MICHELL

MÁY ĐO GIÓ

NO. 7002

NO. 7604-2


NO.7753-2
m/s
m/s



No. 3372-00 AIR VELOCITY TRANSMITTER MODEL SK-ATV

SPECIFICATIONS

Measurement object : Clean air at normal pressure and humidity
Measuring range : 0 to 30 m/s
Analog output : 0 to 300 mVDC / 0 to 30 m/s (linear output)
Accuracy : ±3% F.S.
Temperature compensation: 0 to +60℃
Condition for measurement
- Ambient temperature for main body : +5 to +45℃
- Ambient humidity for main body : Less than 85%RH
- Ambient temperature for sensor : Within th erange fo temperature compen-sation (0 to 60℃)

Power requirements : 12 VDC
Power consumption : 200 mA (Max.)
Outer dimensions : 90 (W) x 50 (H) x 190 (D) mm
Weight : Approx. 400 g
Sensor
- Sensor element : Constant temperature differential Platinum
- Outer dimension : φ8 x 200 mm
- Cord length : φ5.5 x 5 m

Standard accessories : Fitting with screws for probe

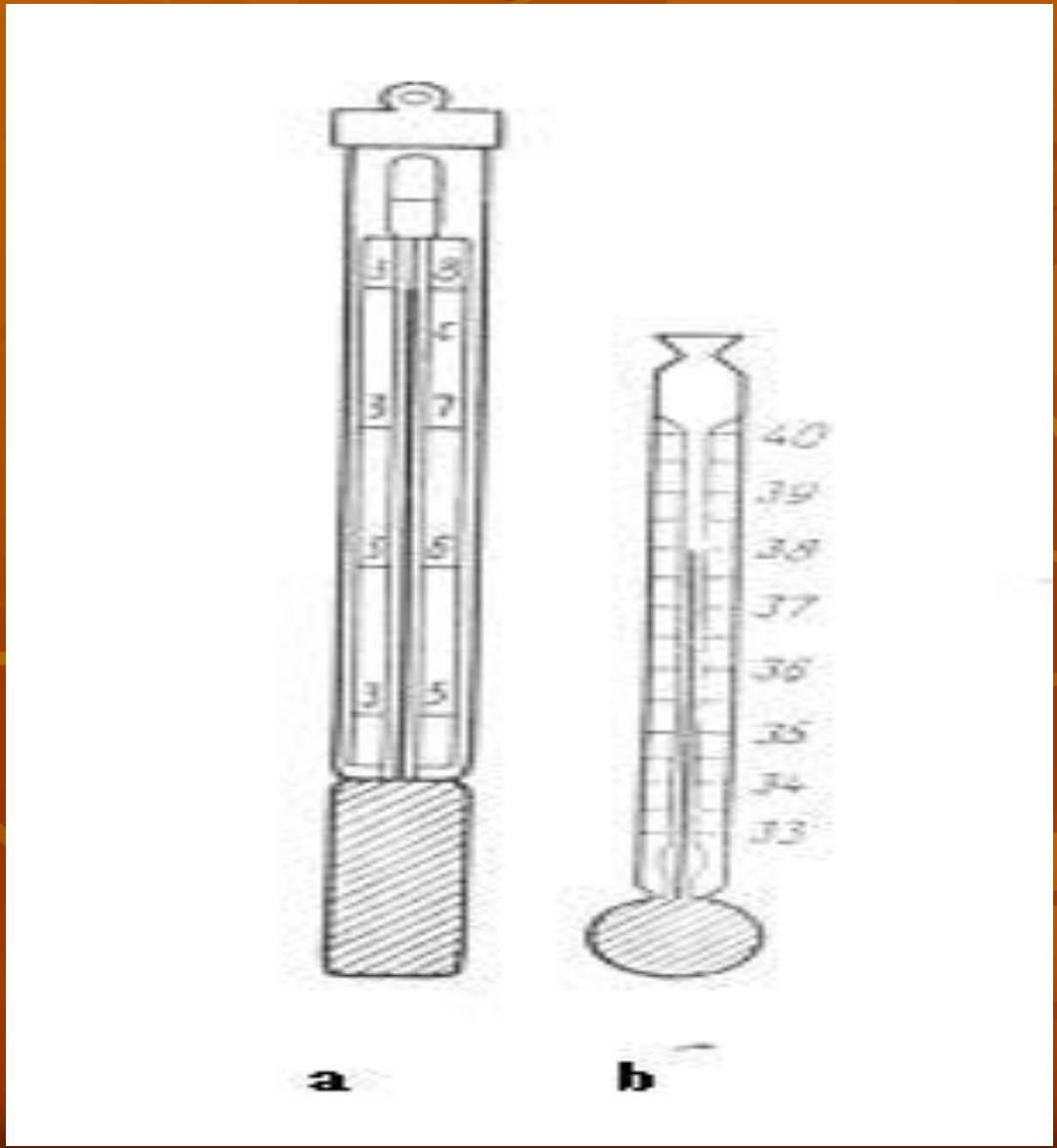
Starhead
production
323
68.7

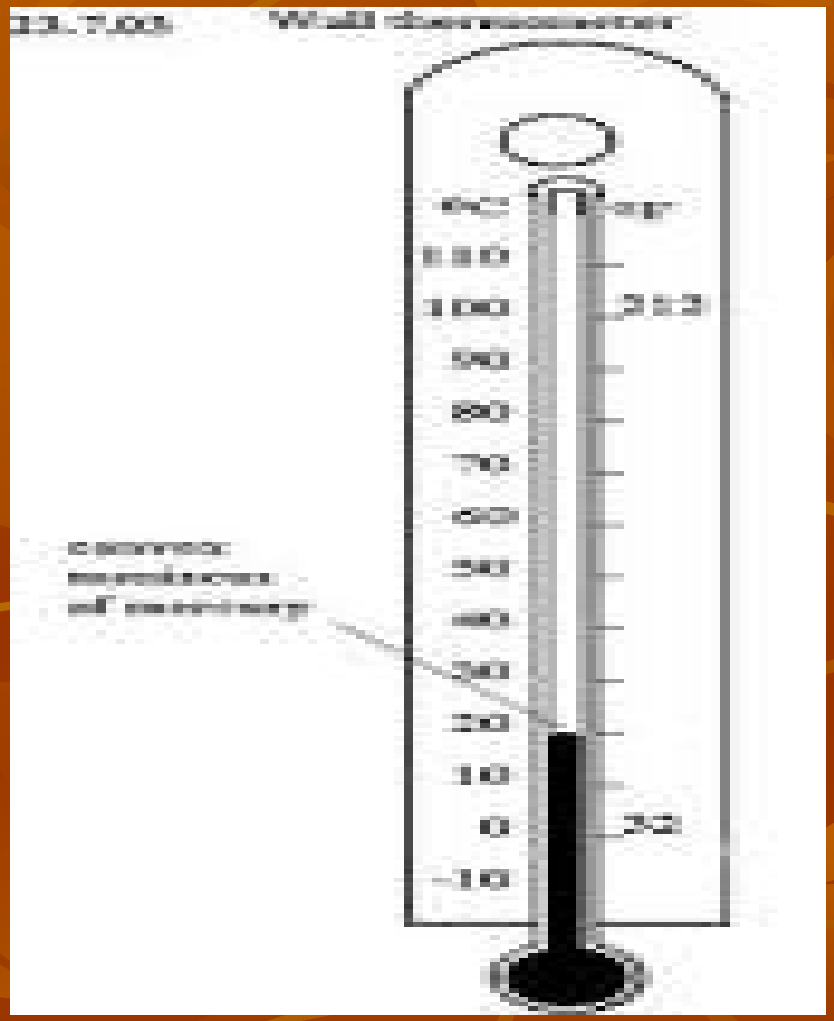
0.0

MultiMeasure
PROFESSIONAL
311
222

SATO
ANEMO-TRANSMITTER
SX-ATV

GIẢM NHIỆT KẾ CATA



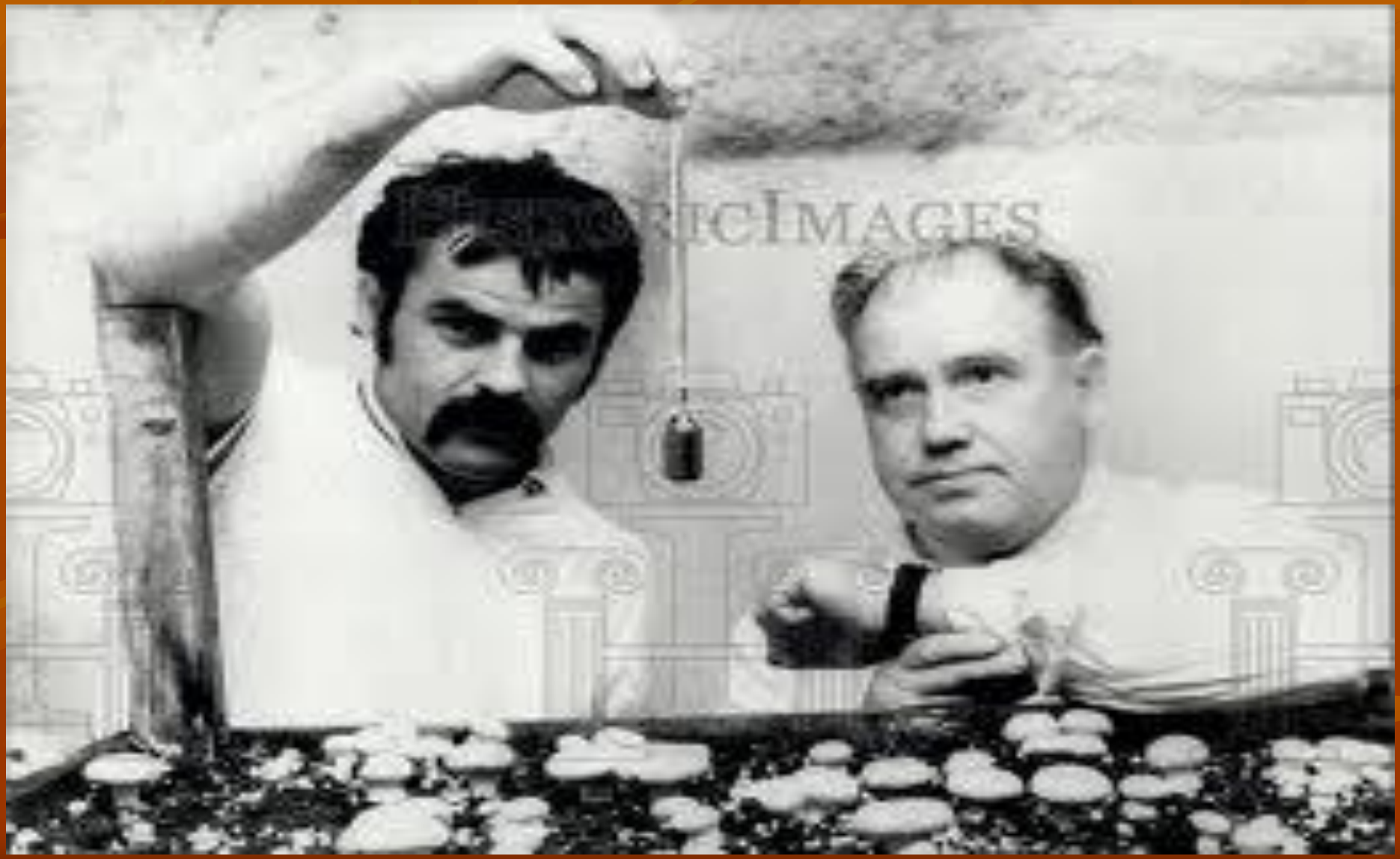

°C
°F
110
100
212
90
80
70
60
50
40
30
20
10
0
32
-10

GIẢM NHIỆT KẾ CATA

SỨC LÀM LẠNH CỦA KHÔNG KHÍ

$$H_o = \frac{F}{T}$$

$$V = \left(\frac{\frac{H}{Q} - 0.20}{0.40} \right)^2$$

$$V = \left(\frac{\frac{H}{Q} - 0.13}{0.47} \right)^2$$

SỬ DỤNG GIẢM NHIỆT KẾ CATA